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Yield and Yield Regulation



Yield: The volume or number of stems that can be removed annually or periodically or the area over which fellings may pass annually or periodically, consistent with the attainment of objects of management.

Yeild Regulation: A term generally applied to determination of yeild and prescribed means of realizing it.

It means fixing in advance, usually for a short period (working Plan period) the amount of timber or other produce which may be removed from the forest annually or periodically.

Types of Yield

Intermittent yield — Thinnings

Final yield — Final yield.

Basis and Methods of Yield Regulation

Basic of yield regulation: No recognised Correlation between a method of yield regulation and a Silvicultural System

Methods of Yield Regulations

Area Method

Volume Methods

Use of formulae

By Number & Size of tree

Computer Simulations

Area Method

The area method says that, in the case of an even aged forest, the area that should be harvested every year is equal to the total area divided by the rotation period.

$$A(h) = A(t)/R$$

where $A(h)$: Area harvested each ayear

$A(t)$: Total area of the Stand

R: Rotation Period

Suppose the forest area is 160 ha and the rotation period is 16 ayears. Calculate the area harvested each ayear

$$A(t) = 160\ ha$$

$$R = 16\ yrs$$

$$A(h) = \frac{160\ ha}{16}$$

$$= 10\ ha/yr$$

Advantages of area method

- Simple
- Works for even-aged crop
- Each coupe can be intensively treated [Attention due to same age]

Disadvantages

- Difficult to predict volume
- Not suitable for uneven-aged crop.

Volume Method

It says that the volume that should be harvested every year is equal to the total volume divided by the rotation period.

$$V_y = \frac{\overline{Tv}}{R}$$

Where V_y = Volume per year

$\overline{Tv}$ = Total volume

R = Rotation period.

Advantage

Simple.

4 Disadvantages

- Does not include growth and increment
- Does not talk about the site quality

Hundeshagen's Method of Yield Regulation

Based on the notion that you can cut more when you have more inventory than a target normal forest and you can cut less when you have less inventory than a target normal forest.

Harvest (Ht) $\propto$ Inventory (It)

N B Curve

